

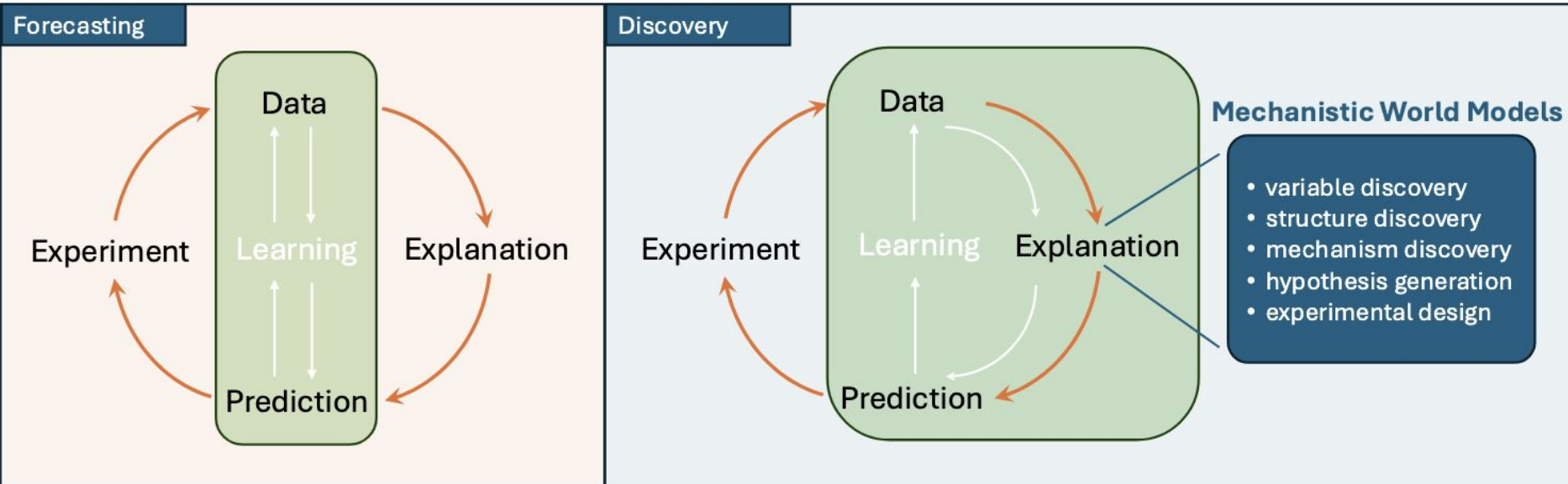
World models

Mechanistic world models

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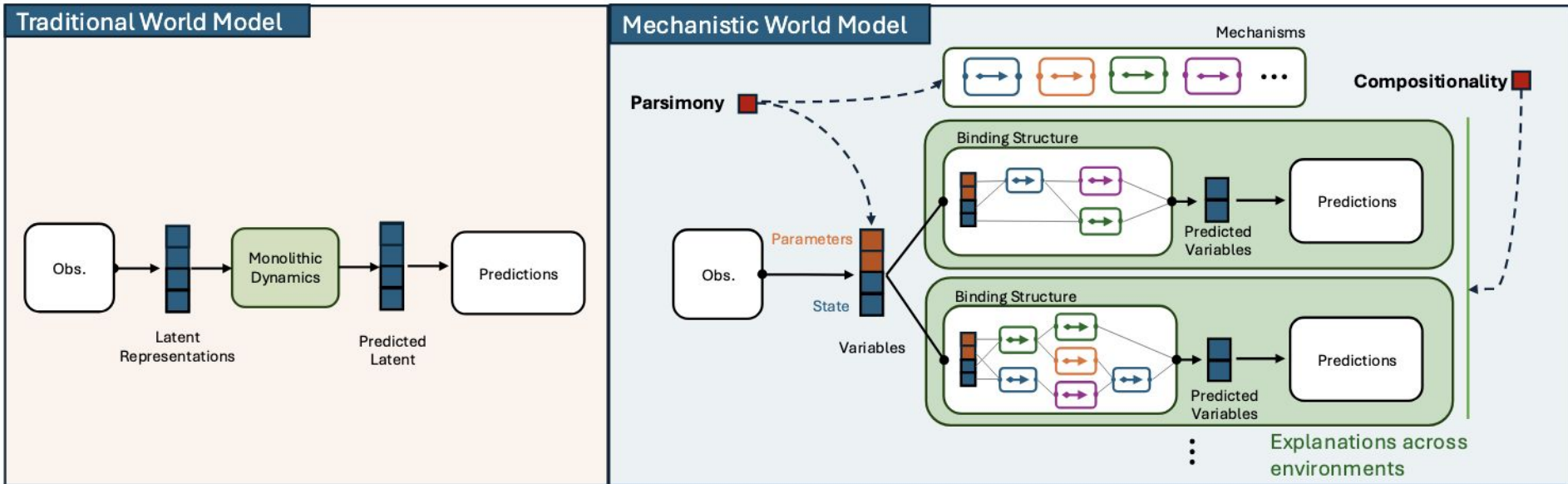
What is world model

- a model that learns to latent environment states from observations and uses them for prediction



What is *mechanistic* world model

- More modularity compared to world models
- prediction loss + **parsimony** + **compositionality**



What is *mechanistic* world model

$$\mathcal{W} = (\underbrace{\mathcal{T}}_{\text{types}}, \underbrace{Z}_{\text{variables}}, \underbrace{\mathcal{M}}_{\text{mechanisms}}, \underbrace{S}_{\text{structure}})$$

Component	Notation	What it is
Types	$\mathcal{T} = \{\tau_1, \tau_2, \dots\}$	Learned functional roles — analogous to <i>position, velocity, mass</i> .
Variables	$Z = \{z_1, \dots, z_n\}, T : Z \rightarrow \mathcal{T}$	This system's latent state plus the latent parameters governing its dynamics, each assigned a type
Mechanisms	$\mathcal{M} = \{(\Sigma_m, f_m)\}$	A typed signature plus a transformation
Structure	$S = \{b_i\}, b : \Sigma_m \rightarrow Z$	Bindings that fill a mechanism's abstract roles with this system's actual variables

What is *mechanistic* world model

Parsimony: account for the most observations with the fewest underlying principles.

Compositionality: build new explanations by recombining existing ones rather than learning fresh ones.

$$\mathcal{W} = (\underbrace{\mathcal{T}}_{\text{types}}, \underbrace{Z}_{\text{variables}}, \underbrace{\mathcal{M}}_{\text{mechanisms}}, \underbrace{S}_{\text{structure}})$$

$$\mathcal{W}^* = \arg \min_{\mathcal{T}, Z, \mathcal{M}, \{S_e\}_{e \in \mathcal{E}}} \left[\Omega_p(\mathcal{T}, \mathcal{M}, Z) + \Omega_c(\{S_e\}) + \sum_{e \in \mathcal{E}} \mathcal{L}_{\text{pred}}(\mathcal{T}, Z, \mathcal{M}, S_e) \right]$$

Things related to mechanistic world models

1. Mechanistic Interpretability
2. Causality and Representation Learning
3. Equation Discovery
4. Information Organisation via Mixtures of Experts